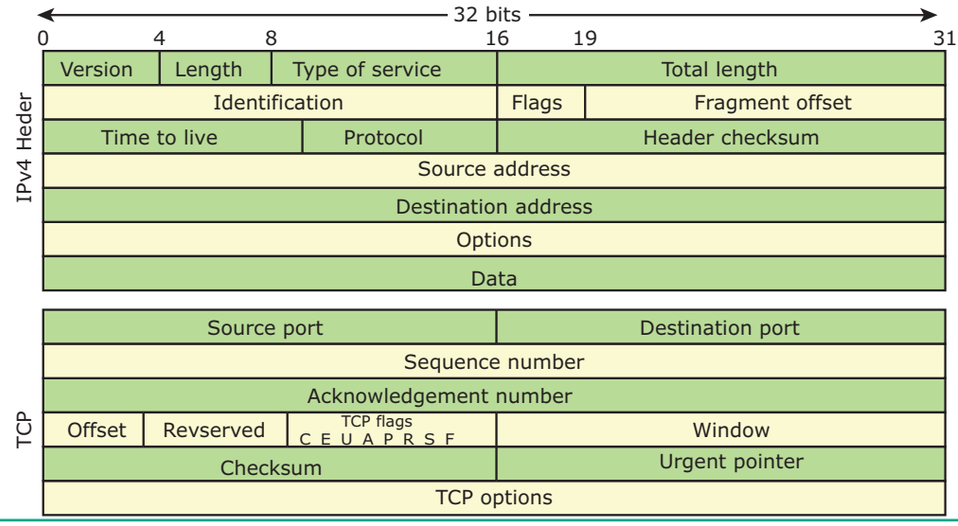


Figure 11.10 TCP/IP Layer

- **HTTPS** – It is used between a web client and a web server ensures *secure* data transmissions.
- **FTP** – It is used between computers for sending and receiving file.

Domain Names and TCP/IP Addresses

The address for any website is not as easy as to remember, domain name are used instead. For example, **216.58.216.164** is one of the IP address for Google and **google.com** is the domain name.

The Different Layers of TCP/IP

There are four total layers of TCP/IP protocol, each of which is listed below with a brief description.

- **Network Access Layer** - concerned with building packets.
- **Internet Layer** - describes how packets are to be delivered.
- **Transport Layer** - ensure the proper transmission of data.
- **Application Layer** - application network processes. These processes include File Transfer Protocol (FTP), Hypertext Transfer Protocol (HTTP), and Simple Mail Transfer Protocol (SMTP).

11.2.3 Other Network Protocols

Network protocols other than OSI and TCP/IP were simply known as other network protocols which implements security over the network communication that include **HTTPs, SSL,** and **SFTP**. Other networks similarly classified in network layer are **IP, ARP, ICMP,IGMP,** at transport layer are **TCP,UDP** at Application Layer are **HTTP, FTP, Telenet, SMTP,** and **DNS.**

HTTPS positions for Hypertext Transfer Protocol Secure. It's a protocol where encoded data transfer on a secure connection. This HTTPS make data more safe and provides data security over the network mainly on public networks like Wi-Fi. See Figure 11.11

For example, let us take a bank website, when we go to login page, we may watch an HTTPS in address bar with some specific design. HTTPS mainly deals with financial transactions or transfer users personal data highly sensitivily. Banking websites are common examples for HTTPS. Data exchanged between the user and the website is not stolen, read or altered by a third party.

XII CompApp_English version CHAPTER 11.indd 156 23-02-2020 12:39:10